Hsiang (Sean) Fu

(737) 274-6945 | hsiangfu@asu.edu | LinkedIn | Website | GitHub

Education

Ph.D. in Computer Science Aug 2025 – Present
Arizona State University GPA: 4.0/4.0

B.S. in Informatics, Human-Centered Data Science Aug 2022 – Dec 2024
The University of Texas at Austin GPA: 3.94/4.0

Research Experience

Graduate Research Assistant Aug 2025 – Present
Interactive Robotics Laboratory, Arizona State University Advisor: Dr. Heni Ben Amor

- Develop language- and vision-conditioned learning methods for robot manipulation, with an emphasis on policy search, reward estimation, and adaptive robot behavior.
- Build and evaluate research systems across UR robots, dexterous-hand hardware, and MuJoCo-based bimanual manipulation environments.

Undergraduate Research Assistant Jan 2023 – May 2024
School of Information, The University of Texas at Austin Advisor: Dr. Abhijit Mishra

- Co-developed ReVision, a compact ~250M-parameter vision-language model and a dataset of 39,000+ image–instruction pairs across 14+ domains for privacy-preserving visual instruction rewriting.
- Demonstrated that quantized models under 500 MB can support task-oriented multimodal instruction processing on resource-constrained devices; work accepted at IJCNLP–AACL 2025.

Selected Research Projects

Honda Dexterous Hand Project Aug 2026 – Present

- Built a manipulation-data pipeline that synchronizes nine camera streams with joint states, force-control signals, and tactile observations.
- Implemented an in-context reinforcement learning framework for adapting dexterous-hand policies.

ComposeR: Training-Free Reward Estimation for Robot Manipulation (Submitted to ICRA 2027) Jan 2026 – Present

- Developed a training-free framework that decomposes language-conditioned manipulation tasks and uses VLM token probabilities to estimate dense, interpretable progress rewards from trajectory videos.
- Combined multi-view evidence, verification passes, and temporal-dependency checks to achieve 92.35% task-outcome prediction accuracy, outperforming evaluated baselines.

UR5 Golf: In-Context Policy Search for Robot Putting Jan 2026 – May 2026

- Built an in-context reinforcement learning pipeline for a UR5 minigolf task, integrating physical rollouts with camera-based outcome feedback.
- Developed a trajectory-optimization workflow using Dynamic Movement Primitive fitting, automated rollout execution, and global search over DMP parameters.

Bimanual Manipulation with Demonstration-Guided Policy Search Aug 2025 – Present

- Built a MuJoCo policy-search pipeline that converts teleoperated demonstrations into parameterized 3-D wrist trajectories using Dynamic Movement Primitives.
- Integrated LLM-generated DMP proposals, simulator evaluation, reward shaping, and experiment logging to automate iterative policy optimization.

Introduction to In-Context Learning [Project] Aug 2025 – Present

- Built an educational website with interactive demonstrations of in-context learning across sequence prediction, regression, classification, and reinforcement learning, reaching 5,000+ views within six months.

Selected Publications

- A. Mishra, **H. Fu**, M. Li, *et al.* "ReVision: A Dataset and Baseline VLM for Privacy-Preserving Task-Oriented Visual Instruction Rewriting." *IJCNLP–AACL*, 2025. **[Paper]**

Technical Skills

Languages: Python, Java, SQL

Machine Learning: PyTorch, reinforcement learning, vision-language models, large language models, in-context learning

Robotics: MuJoCo, UR5 and UR robots, Dynamic Movement Primitives, teleoperation, trajectory optimization, multi-camera data collection

Tools: Git, Linux, scikit-learn

Teaching Experience

Graduate Teaching Assistant Aug 2025 – May 2026
CSE 101 and CSE 301, Arizona State University

- Led three weekly laboratory sections covering programming fundamentals and ethical computing, providing hands-on instruction and technical guidance.